Casey Barton

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PROFILE

I build and operate agent-powered features end to end — model orchestration, agent capabilities and integrations, reliable asynchronous workflows for audio, image, and video generation — with the eval, monitoring, and testing discipline that keeps a live product trustworthy. Every agent I ship runs under an eval gate first: golden sets, behavioral assertions, judge criteria, full trace logging.

Three years as a founder running an evidence-grade agent control plane from Honolulu: hand-rolled multi-agent loops (planning, tool-use, retrieval, memory, guardrails, human handoffs); **65** MCP connectors exposing governed context to any MCP-compatible system; generative-media agents over fal.ai's 600+ model catalogue (text-to-image, image-to-video, audio, queued async endpoints); an offline faster-whisper transcription pipeline; model-agnostic abstractions with per-task latency and token-cost benchmarking on every release.

CORE COMPETENCIES

Engineering: TypeScript · Python · Node.js · SQL · Bash · REST APIs · containers · GitHub Actions (CI/CD) · Supabase · Postgres · event-driven systems · async job queues · observability & run logging

AI & Creative Agents: production agents · model orchestration · MCP servers & connectors · generative media (audio/image/video) · LLM evaluation harnesses · judge-in-the-loop patterns · trace intelligence · failure-mode instrumentation · RAG · function calling · structured outputs · guardrails · human handoffs

Production: full-stack ownership (discovery → deploy → operate) · async workflows with retries & durable state · eval/monitoring/testing · CI regression gates · decision audit receipts · cost/latency tradeoffs

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Own agent-powered features from discovery and prototyping through deployment, monitoring, and iteration: model orchestration, agent capabilities and integrations, reliable asynchronous workflows — fail-closed by default, customer feedback driving what gets built next.
- Build generative-media agents: orchestrated image, video, and audio generation over fal.ai's 600+ model catalogue — queued async endpoints with durable job state, progress visibility, and evaluation gates on media output — plus a fully offline faster-whisper transcription pipeline (WhisperX lineage, isolated venv, pinned deps, idempotent batch processing).
- Ship the debugging story they ask about: trace every run through structured logs, job state, and per-task latency instrumentation; diagnose queue saturation, retry behavior, provider latency, and worker defects — then fix the root cause and add the alert so it cannot recur.
- Own the harness: eval suites, traces, and judge-in-the-loop patterns that tell whether an agent is actually doing the job, against a baseline. Prompt and configuration changes are validated by eval harnesses before anything touches production — an agent that cannot be measured cannot be trusted.

- Author **65** MCP connectors and integrations that expose tools and structured resources to any MCP-compatible system — the shared, governed context layer: ERPs, CRMs, data pipelines, webhooks, messaging — with evidence boundaries and audit receipts.
- Run an evidence-grade agent control plane across a **1,413-repository** estate: planning and tool-use loops, retrieval over a domain ontology, memory, guardrails, typed handoff contracts, human approval gates on consequential actions — backed by a **989-test** suite, **95/95** verification checks green.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — evaluation-gated production pipeline

- Attributable hourly production pipeline with immutable evidence models, stable finding codes, and fail-closed **GO / NO-GO** decisions; recovery restricted to contingency evidence declared before reassessment.
- Verification gates on every agent action: no agent completion claim without independent provider readback — the eval gate decides, not the agent.

GlacierEQ estate — production agent systems

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Per-task latency instrumentation and decision audit logs across agent runs — the tracing practice that makes debugging and benchmarking honest.

SKILLS

Languages: TypeScript · Python · SQL · Bash · JavaScript / Node.js

AI / Creative Agents: production agents · model orchestration · MCP integrations · generative media pipelines · eval harnesses · judge-in-the-loop · trace intelligence · failure-mode instrumentation · benchmarking · memory systems · guardrails · typed handoff contracts

Production: full-stack engineering · async workflows · observability & run logging · CI regression gates · fail-closed recovery · cost/latency tradeoffs · documentation & handover

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment, or certification.